

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 18:29

File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	llama3.2

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by an artificial intelligence system using Phi-3 Mini as the language model. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Editorial Decision

MINOR REVISION

Weighted Score: 3.6 / 5.0
Confidence: 77%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.0	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	2.5	Weak	■■■■■
Statistical Analysis	4.2	Good	■■■■■
Results & Evidence	4.0	Good	■■■■■
Discussion & Conclusions	4.0	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	2.5	Weak	■■■■■
References & Citations	4.2	Good	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Weaknesses:

- [TITLE] could be more concise

Major Comments:

- [ABSTRACT] should not contain references
-

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Weaknesses:

- Transition between sections could be improved
- Section balance is not perfect

Major Comments:

- Consider adding a clear statement of the research problem in the introduction

Recommendations:

- Add an abstract section to summarize the main contributions and findings
 - Provide more details on the preprocessing steps applied to the measured signals
 - Improve section balance by reorganizing or removing sections as needed
-

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 2.5 / 5.0 | Confidence: 55%

Weaknesses:

- The logical flow could be improved by presenting the methodology before the introduction.

Recommendations:

- Consider reordering the sections to improve logical flow.
-

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 4.2 / 5.0 | Confidence: 80%

Strengths:

- Bootstrap confidence intervals and Holm correction applied correctly

Weaknesses:

- Kruskal-Wallis test may not be the best choice for this data type
- Assumption of normality verified, but non-parametric methods justified when used

Major Comments:

- The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- The use of Pareto indicators and hypervolume is not clearly justified

Recommendations:

- Consider using non-parametric methods when dealing with non-normal data
- Verify homoscedasticity assumption before applying Kruskal-Wallis test

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.0 / 5.0** | Confidence: 62%

Strengths:

- 6 figure references were detected in the manuscript.
- 10 table references were detected in the manuscript.
- 52 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the problem.

Weaknesses:

- Some results are not clearly tied to figures or tables, making it difficult to assess their accuracy.

Major Comments:

- The manuscript could benefit from more detailed explanations of the preprocessing steps and how they impact the results.

Recommendations:

- Provide more detailed explanations of the preprocessing steps and their impact on the results.
- Include more figures or tables to support the quantitative results and facilitate assessment of accuracy.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **4.0 / 5.0** | Confidence: 90%

Weaknesses:

- The discussion could benefit from more explicit comparison with prior work in the field of structural parameter identification.

Major Comments:

- The Discussion section could be improved by providing more context on the limitations of classical metaheuristic approaches and how EANN addresses these limitations.

- More emphasis should be placed on the theoretical implications of using EANN for structural parameter identification.

Recommendations:

- Consider adding a section to discuss the potential applications and challenges of using EANN in real-world scenarios.
 - Provide more details on how the performance metrics (RMSE, MAE, R2) were chosen and validated.
-

WritingReviewer

Scientific writing quality, clarity, and style

Score: **2.5 / 5.0** | Confidence: 80%

Weaknesses:

- type: overused opener; sentence: This paper proposes a novel...
- type: run-on sentence; sentence: Modeling and identification of structural parameters are fundamental to control-oriented civil engineering applications, including structural health monitoring, active, semi-active, and passive vibration control for seismic response mitigation, and state estimation. In numerous studies, structural deformations are effectively represented using lumped mass-spring-damper models [1], [2] which are also widely used in damage detection [3] and observ

Major Comments:

- Consider rephrasing sentences to improve clarity and concision
- Define acronyms consistently throughout the manuscript

Recommendations:

- Use transitional phrases to connect ideas between paragraphs
 - Provide definitions for technical terms on first use
-

ReferencesReviewer

References quality, recency, and relevance

Score: **4.2 / 5.0** | Confidence: 80%

Strengths:

- A good mix of recent references (2021-2022) and seminal works (1998-2000)

Major Comments:

- The references section could benefit from more diverse sources, including academic journals other than IEEE and Korea-Australia Rheology Journal.

Recommendations:

- Consider adding more recent references to the list of seminal works or to provide a more comprehensive literature review.
-

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [ABSTRACT] should not contain references
- [2] Consider adding a clear statement of the research problem in the introduction
- [3] The manuscript could benefit from a more detailed discussion on the assumptions of the Kruskal-Wallis test
- [4] The use of Pareto indicators and hypervolume is not clearly justified
- [5] The manuscript could benefit from more detailed explanations of the preprocessing steps and how they impact the results.
- [6] The Discussion section could be improved by providing more context on the limitations of classical metaheuristic approaches and how EANN addresses these limitations.
- [7] More emphasis should be placed on the theoretical implications of using EANN for structural parameter identification.
- [8] Consider rephrasing sentences to improve clarity and concision
- [9] Define acronyms consistently throughout the manuscript
- [10] The references section could benefit from more diverse sources, including academic journals other than IEEE and Korea-Australia Rheology Journal.
- [11] Conflict-of-interest declaration was not detected.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Specific Improvement Recommendations

1. Add an abstract section to summarize the main contributions and findings
2. Provide more details on the preprocessing steps applied to the measured signals
3. Improve section balance by reorganizing or removing sections as needed
4. Consider reordering the sections to improve logical flow.
5. Consider using non-parametric methods when dealing with non-normal data
6. Verify homoscedasticity assumption before applying Kruskal-Wallis test
7. Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
8. Provide more detailed explanations of the preprocessing steps and their impact on the results.
9. Include more figures or tables to support the quantitative results and facilitate assessment of accuracy.
10. Consider adding a section to discuss the potential applications and challenges of using EANN in real-world scenarios.
11. Provide more details on how the performance metrics (RMSE, MAE, R2) were chosen and validated.
12. Use transitional phrases to connect ideas between paragraphs